ITA06020 MACHINE LEARNING

Experiment 10 – Implementation of Expectation Maximization (EM) Algorithm

Aim:

To implement the Expectation Maximization (EM) algorithm in Python to group data into clusters using a real-world dataset.

Dataset: Iris Flower Dataset (available in Scikit-learn).

The EM algorithm is implemented using the Gaussian Mixture Model (GMM).

Procedure

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Select the input features.
4. Create a Gaussian Mixture Model with three clusters.
5. Fit the model using the EM algorithm.
6. Predict the cluster labels.
7. Display the predicted clusters and model results

Python code:

```
from sklearn.datasets import load_iris

from sklearn.mixture import GaussianMixture

data = load_iris()

X = data.data

model = GaussianMixture(n_components=3, random_state=42)

model.fit(X)

clusters = model.predict(X)

print("Expectation Maximization Algorithm")

print("Total Samples:", len(X))

print("Number of Clusters:", 3)

print("\nFirst 10 Cluster Predictions:")

print(clusters[:10])

print("\nCluster Distribution:")

for i in range(3):
```

print("Cluster", i, ":", sum(clusters == **i), "samples")**

Output:

```
        print("Cluster", i, ":", sum(clusters == i), "samples")
```

```
Expectation Maximization Algorithm
Total Samples: 150
Number of Clusters: 3

First 10 Cluster Predictions:
[1 1 1 1 1 1 1 1 1 1]

Cluster Distribution:
Cluster 0 : 55 samples
Cluster 1 : 50 samples
Cluster 2 : 45 samples
```

Result

Thus, the Expectation Maximization algorithm was successfully implemented using the Gaussian Mixture Model on the Iris dataset. The dataset was grouped into three clusters.